



FirstDrop

The warning you need, in the language you speak.

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FirstDrop

Flood warning in your language,
at your level.

“I need a localized warning,
If it's a physical alarm, I see it,
and I hear it.”
– Nancy, Brooklyn



THE PROBLEM

Basement residents face life-threatening risks because current systems fail to deliver the right warning, in time, at their level.



Delayed, non-localized city alerts

Citywide notices are too late and too broad to be actionable.



English-only emergency communication

Critical information isn't always accessible for non-English speakers.



High uncertainty when away or asleep

Residents can't monitor their home in real time or know what to do in a flood.



RESULT: Preventable damage, panic, and safety hazards.



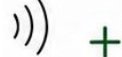
OUR SOLUTION

A dual system that detects flooding instantly and delivers life-saving instructions in your language.



PHYSICAL SENSOR

Moisture probes detect water instantly and trigger a loud physical alarm.



MOBILE APP

Step-by-step survival instructions in your primary language.

HOW IT WORKS



CORE VALUE



Hyper-local accuracy
In-home detection eliminates false alarms.



Zero translation lag
Instructions in your language instantly.



Accessible reliability
Physical alarm + mobile alerts ensure you hear and act.

IMPACT



Reduces fatal risks



Enables faster response and prevention



Protects property and important documents



Bridges language and infrastructure gaps in emergency systems

WHO IT SERVES



Basement apartment residents (NYC)



Non-English / ESL speakers



Landlords & Building Owners



Community Organizations



First Responders

KEY STAKEHOLDERS



BOTTOM LINE

FirstDrop combines in-home detection with a multilingual app to deliver the right warning, at the right time, in the language you speak—so you can act fast and stay safe.



Low-cost device
Target price: \$15–\$20

Target Users + Other Key Stakeholders

Target Users



Non-English or
English as a Second
Language Speakers



NYC Basement
Apartment Residents

Secondary Stakeholders



Landlords



Building Owners



Community Hubs



First Responders

Problem: Insights Into Users and Stakeholders

The Core Problem: Systemic infrastructure failures and delayed, English-only city alerts leave vulnerable basement residents completely exposed to immediate, life-threatening hazards.

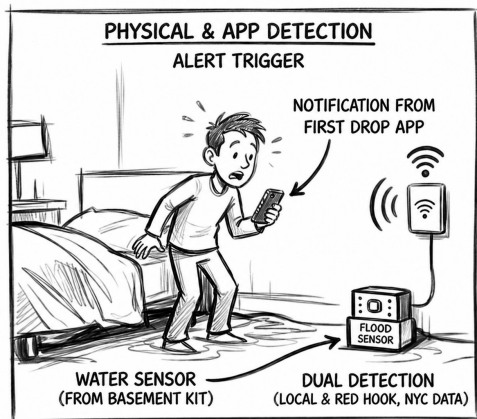
"Inefficient sewers flooded my home. The biggest risk was electrocution."

–Ray, Queens

"I need a localized warning. If it's a physical alarm, I see it, and I hear it."

–Nancy, Brooklyn

Solution and Value to Users/Stakeholders



Hyper-Local Accuracy: In-home sensor eliminates false alarms by detecting water precisely at the unit level.



Zero Translation Lag: Mobile app instantly delivers step-by-step survival instructions in the user's primary language.

Accessible Reliability: Bridges the gap between a low-tech physical alarm (for immediate waking/alerting) and a smart digital interface (for remote monitoring and protocols).

Pre-Disaster Experience

Residents in NYC's basement units currently manage a constant mental burden during the rain. Without data specific to their individual floor level, they must rely on broad weather forecasts and physical observation.



Preventative Labor

When a storm is forecasted, residents often perform manual maintenance, such as clearing debris from outdoor drains by hand or with makeshift tools.



Property Preparation

Elevating electronics and personal items onto cardboard boxes or higher surfaces to avoid floor-level water.



Information Filtering

Relying on slow community translations or social media because official city alerts are generic and English-only.

During & Post-Disaster Experience



Detection Gap

Missing unit-level alerts while away or asleep, leading to avoidable damage to furniture and immigration records.



Linguistic Confusion

Handling basement hazards like submerged outlets or pressurized doors without translated emergency protocols.



Remote Anxiety

Facing home uncertainty while away, preventing early actions like notifying neighbors or landlords to check the unit.



Recovery Burden

Managing high replacement costs and complex insurance negotiations without factual evidence of when water entered.

**During
Disaster**

**Post
Disaster**

How It Works

01

The user installs the sensor by placing on the ground of their basement area in a flood prone area, downloading the app and setting up location + language settings prior to the flood event.

02

The physical sensor detects water by looking for when the electrical probes are completely submerged in water which completes an electrical circuit.

03

The sensor then sends a message via a wifi connection with immediate instructions, it also prompts the user to open the FirstDrop App.

04

Based on user interaction and information from the user, the app will provide step by step instructions to protect the user from dangerous scenarios.

05

If a flooding event triggers several sensors in a singular neighborhood, any user registered to that area will receive a potential flood notification.

Note: Users without the sensor will be able to access the step-by-step instructions but the whole process must be manually followed unless community alerts were activated.

Solution Ecosystem



Hardware Layer

Moisture-sensitive floor sensor triggers a local audio alarm independently of the internet.



Data Layer

App integrates broad neighborhood flood data with a dedicated feed for the user's specific unit.



Linguistic Layer

Uses geolocation to default to primary community languages, such as Chinese in Flushing.



Human Layer

Sensor distribution and setup support through local churches and community hubs to reach older demographics.



Stakeholder Loop

Opt-in alert sharing with landlords to accelerate response times for building-wide mechanical failures.

Table of Comparison to Alternatives

	FirstDrop	NYC Emergency Notifications	Existing Water Sensors	LinkNYC
Hyper -Localized Warnings	✓	✗	✓	✗
Sent In Native Language	✓	✓	Depends	✗
Provides Survival Instructions	✓	✗	✗	✓
Cost	\$15-\$20	Free	\$15-\$65	Free

FirstDrop is not only a water sensor but a tool to help those who don't speak English stay safe in flood events.

Fairness: How Does the System Deliver Fairly

- The goal is to use existing technology to keep the price of the alarm reasonable
- Hope is to partner with local non-profits and look for the government to support subsidize the price of the device for highly impacted neighborhoods
- Those with the FirstDrop App will get flood notifications if several sensors go off in a certain radius
- Learning aspect of the FirstDrop is available via the FirstDrop App
 - Goal is for the app to be a *learning resource* in its own right

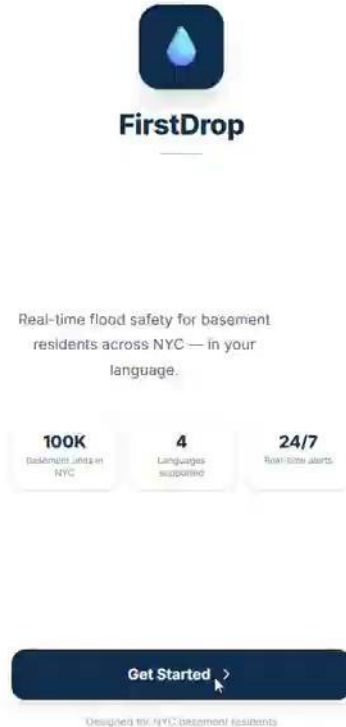
Elevator Pitch

Imagine waking up to rising water in your basement apartment. Panic sets in because official city alerts are notoriously late and rarely in your native language.

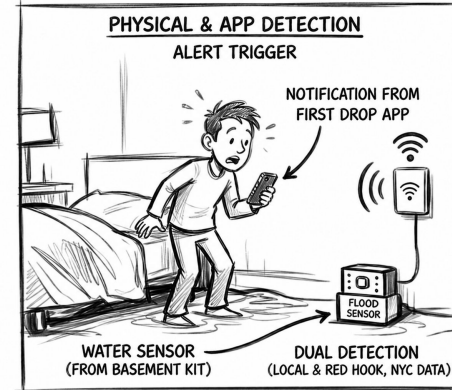
FirstDrop solves this by pairing a physical, in-home water sensor with a neighborhood-adaptive app. The second water breaches your floor, the sensor triggers a loud physical alarm and instantly pushes step-by-step survival protocols to your phone in your primary language.

Whether you're home asleep or away at work, FirstDrop replaces translation delays with the immediate, hyper-local guidance you need to protect your life and property.

Prototype



FirstDrop App



Sensor with Scenario